OLLSCOIL TEICNEOLAÍOCHTA BHAILE ÁTHA CLIATH
TU DUBLIN
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STABILITY SELECTION IN LOGISTIC REGRESSION

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INTRODUCTION

- Transition from Primary to Secondary School
  - Growing up in Ireland Study
- Project Objectives
  - Predictor-response variables and Logistic Regression Model
  - Monte Carlo Simulations & Inferential Statistics
  - Traditional Model Selection (Step wise & Lasso)
  - Advanced Model Selection via Stability Selection
  - Comparative study on model selection methods

SOFTWARES USED

- Software used for this project are :

  - R language

    Major libraries and functions :

    1. *glm*
    2. *glmnet*
    3. *step*
    4. *stabs*

  - Microsoft Word
  - PowerPoint

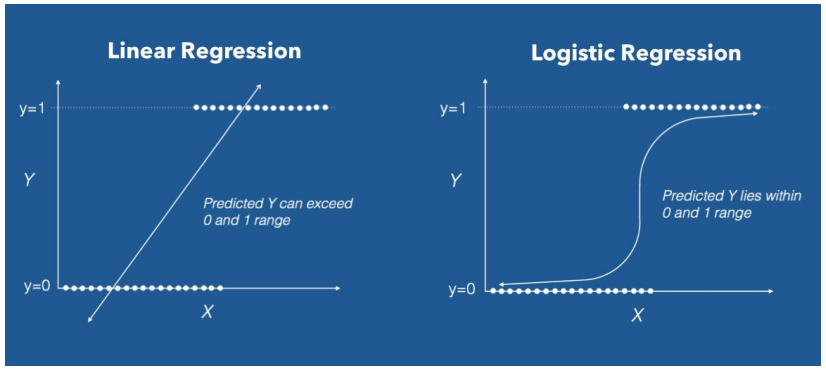




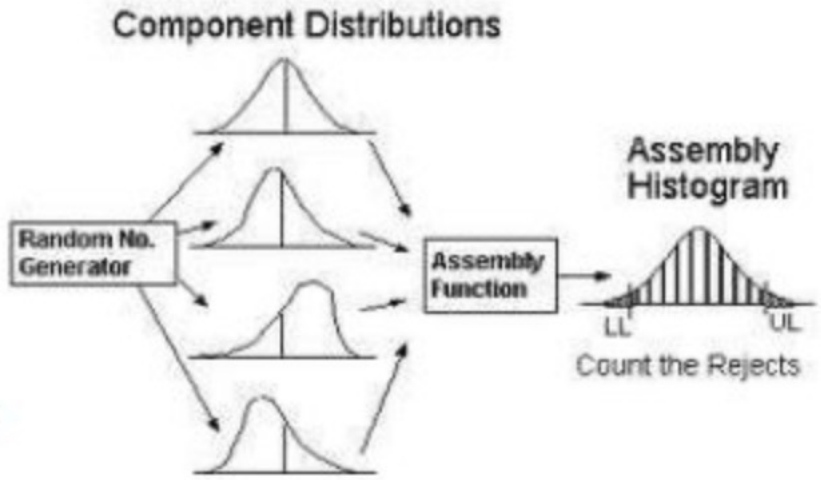


LOGISTIC REGRESSION & MONTE CARLO SIMULATIONS

- Logistic Regression:
  - Description – *Generalized Linear Model Family*
  - Purpose of use – *Binary/Binomial response & Mathematical Formulation*
  - Inference - *Hypothesis Testing*



- Monte Carlo Simulation:
  - Description – *Iterative simulation procedure*
  - Purpose of use – *Decision making under randomness from sampling*



WORK SCHEDULE

- Model Selection : Stepwise and Lasso Regression technique
- Advanced model selection : Stability Selection on Lasso Regression
- Comparison between model selection methods
- Documentation of final project

DECEMBER 2019

APRIL 2020

REFERENCES

- Smyth, E. (2017). Off To a Good Start? Primary School Experiences and the Transition to Second-Level Education: Dublin, The Stationery Office
- Collet, D. (2003). Modelling Binary Data: Boca Raton, CRC Press
- Hosmer, D., Lemeshow, S. (2000). Applied Logistic Regression: New York, John Wiley & Sons, Inc.

QUESTIONS & ANSWERS…